

Mathematical inference

science and logic

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*“I got bored one day and I put everything on a bagel...
everything: All my hopes and dreams, my old report
cards, every breed of dog, every last personal ad on
Craigslist, sesame, poppy seed, salt... and it collapsed
in on itself. ’Cause you see, when you really put
everything on a bagel, it becomes this: The truth.”*

Jobu Topaki in Everything Everywhere All At Once

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CHAPTER 1

Prelude: Andante

“Science, including philosophy, is intrinsically polemical, and the polemical cannot be separated from the agonistic. In epochs in which great new things arise, the agonistic factor usually comes to the fore”

Johan Huizinga in *Homo Ludens*

1. Prelude: Andante

1.1 Inspiration

1.2 Introduction

Scope and assumptions

Research questions

Expected contributions

Relevance

1.3 Outline

1.4 Idiosyncrasies

CHAPTER 2

First movement: Allegro

“Indeed, an insistence that all of mathematics be stated in that language all of the time can place unnecessary burdens on a theory, particularly one intended for application in the real world. It can also degenerate into an affectation, used only linguistically rather than functionally. To give every old, familiar notion a new, impressive name and symbol unknown to Gauss and Cauchy has nothing to do with rigor. It is, more often than not, a form of gamesmanship whose real purpose is to conceal the Mickey Mouse triviality of what is being done. One would blush to state it in plain English”

E. T. Jaynes in *Probability Theory: The Logic of Science*

2. First movement: Allegro

2.1 Solid foundations for mathematics

2.2 Solid foundations for statistics

2.3 A good and honest attitude for doing these sciences

2.4 Mathematical manifesto

CHAPTER 3

Second movement: Andante

“A story is needed before one can set sail. Look at us. For the sake of the story, we are sitting here aboard a water taxi on our way to the Holy Grail, the unknown, and adventure. People think life becomes more enjoyable through stories. That is true enough, but it is not the point. Without stories, it becomes meaningless. Sense consists of sentences. The words that tell the story lay a reassuring relation of cause and effect over random facts and events. People crave plot because a plot reduces the unbearable and unmanageable chaos of the sublunary world to a human scale, and to a chain of initiatives and consequences that a person is able to comprehend. A plot gives a sense of control, provenance and destination, origin and direction”

Ilja Leonard Pfeijffer in *Grand Hotel Europa*

3. Second movement: Andante

3.1 Jaynes' + Cox theory, what is derived

3.2 Objections

Consistency

Reasonableness

Universality

Subjectivity

Causality

CHAPTER 4

Third movement: Scherzo

“A common mistake that people make when trying to design something completely foolproof is to underestimate the ingenuity of complete fools”

Douglas Adams in *Mostly Harmless*

4. Third movement: Scherzo

4.1 Different type of logics

4.2 Type theory

4.3 Martin-Löf-style type theory

CHAPTER 5

Fourth movement: Vivace

*“Scientists have calculated that the chance of anything so
patently absurd actually existing are millions to one.
But magicians have calculated that million-to-one chances crop
up nine times out of ten”*

Terry Pratchett in *Mort*

- 5.1 Context manipulation in Martin-Löf-style type theory**
- 5.2 Modalities inside Martin-Löf-style type theory**
- 5.3 Probability types**
- 5.4 Structural causal models in Martin-Löf-style type theory**
- 5.5 Results**

CHAPTER 6

Fifth movement: Presto

“For far too long economists have sought to define themselves in terms of their supposedly scientific methods. In fact, those methods rely on an immoderate use of mathematical models, which are frequently no more than an excuse for occupying the terrain and masking the vacuity of the content. Too much energy has been and still is being wasted on pure theoretical speculation without a clear specification of the economic facts one is trying to explain or the social and political problems one is trying to resolve”

Thomas Piketty in *Capital in the Twenty-First Century*

Acknowledgements

“Teleutias truly sailed home under the happiest of circumstances. When he went down to the sea to begin his journey home, there was not a soldier who did not clasp his hand. One placed a wreath on his head, another put a fillet around him, and those who came too late, when he was already putting out to sea, still threw wreaths into the water and wished him every good thing.

I realize that I am not describing any great expenditure, any danger, or any strategy worth mentioning; but by Zeus, I do think it is important for a man to consider what kind of conduct enabled Teleutias to evoke such a response from his subordinates, for that is, beyond any amount of money or danger, a man’s most noteworthy achievement”

Xenophon in *Hellenica*

Appendices

APPENDIX A

Definitions

“So if you are an expert and, as I said just now, have deceived me with your promise to give a demonstration about Homer, you are playing a trick on me. But if you are not an expert, but are possessed by Homer through a divine gift and, without knowledge, say many beautiful things about the poet, as I claimed is the case with you, then you are not playing a trick on me. So you need only choose how you wish us to regard you: as a man who plays tricks, or as a man governed by the gods”

Socrates in *Ion*

APPENDIX B

Proofs and arguments

“The important thing is life — life and nothing else! What is any ‘discovery’ whatever compared with the incessant, eternal discovery of life?”

Hippolyte in *Idiot*

B. Proofs and arguments

B.1 Causality argument

B.2 Physics argument

B.3 Excluded-middle proof

B.4 Clayton proof

B.5 SCM-equivalence proof

B.6 Causality-completeness proof

Bibliography/sources

“Contrarily, it is characteristic of a truly wise man that, being just a mortal, he doesn’t want to transgress any boundaries with his wisdom and together with humanity either turns a blind eye, or, out of courtesy, wanders along”

Folly in *The Praise of Folly*

